SR UNIVERSITY / ENGINEERING COLLEGE, WARANGAL
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING | B.TECH CAPSTONE MINI-PROJECT

MediSense NLP: Hospital Service Quality & Regional Diagnostic Report Simplification

A Dual-Engine Healthcare Natural Language Processing Framework & Full-Stack Web Platform

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Academic Year: 2026 - 2027 | Warangal, Telangana

■ PROJECT DOMAIN & BACKGROUND

Transforming Regional Healthcare via Natural Language Processing

Empowering Patients and Administrators in Semi-Urban & Tier-2 District Hospital Ecosystems

Healthcare NLP Domain

• **NLP in Medicine:** Structuring messy clinical text into actionable diagnostic semantics.

• **Unstructured Challenge:** Over 80% of healthcare data resides in unstructured lab sheets & reviews.

• **Patient Paradigm:** Empowering laypatients with clear diagnostic comprehension.

Warangal Healthcare Hub

• **Regional Healthcare Core:** Serves 6 northern Telangana districts with heavy daily footfalls.

• **12 Major Hospitals:** Mix of public medical colleges (MGM, KMC) and private super-specialities.

• **Consultation Strain:** Over 1,500+ daily OPD registrations, restricting doctors to <3 mins.

Dual Healthcare Friction

• **Clinical Jargon Barrier:** Diagnostic reports written in Latinate terminology cause severe anxiety.

• **Feedback Audit Deficit:** Administrations lack tools to isolate urgent grievances from feedback.

• **MediSense Solution:** Dual-engine system addressing both report and feedback bottlenecks.

■ PROBLEM DEFINITION

Critical Gaps in Healthcare Communication & Quality Auditing

Analyzing the Twin Operational Bottlenecks Confronting Outpatients and Hospital Administrators

Problem 1: Diagnostic Lab Report Incomprehensibility

• **Specialized Terminology:** Lab tests contain Latinate pathology terms like 'hypercholesterolemia'.

• **Extreme Readability:** Original lab reports have Flesch-Kincaid Grade Level > 14.8 (post-graduate level).

• **Severely Constrained Doctors:** OPD doctors have < 180 seconds per patient, leaving no time to explain lab sheets.

• **Generative AI Hallucination:** Commercial LLMs alter numeric values (e.g. 142 to 124 mg/dL), introducing life risks.

• **Neglect of Vernacular:** Absence of reliable, verified Telugu explanations and voice audio.

Problem 2: Unstructured Patient Grievance Overload

• **High-Volume Surveys:** Hospitals collect tens of thousands of written reviews without automated triage.

• **Slow Manual Audits:** Clerical staff manually review small survey samples; issues take weeks to surface.

• **Unaddressed Bottlenecks:** Patients face 90-120 minute registration queues, hidden discharge bills, and nurse friction.

• **Lack of Quantifiable Metrics:** Administrations lack an objective Service Quality Index (SQI) to benchmark wards.

• **Isolated Systems:** Clinical report deconstruction and sentiment intelligence operate in complete isolation.

■ RESEARCH MOTIVATION

Why MediSense NLP? The Imperative for Regional Clinical Equity

Addressing Low-Resource Dravidian Demographics and Hospital Operational Efficiency

Pillar 1: Vernacular Equity

• **Low-Resource Dravidian Need:** Over 65% of rural families in northern Telangana are non-English literate.

• **Inequity of Medical English:** Expecting rural citizens to interpret post-graduate reports creates disparities.

• **Telugu Voice Inclusivity:** Telugu voice audio allows illiterate, rural, and elderly patients to understand health autonomously.

Pillar 2: Clinician Efficiency

• **Relieving OPD Load:** Doctors spend significant consultation time repeatedly explaining standard lab sheets.

• **44.7% Time Savings:** Pre-consultation simplified reports reduce explanation overhead from 170s to 94s.

• **Elevated Clinical Care:** Frees physician cognitive focus for physical diagnosis and therapeutic formulation.

Pillar 3: Data-Driven Quality

• **Quantifiable Accountability:** Transforms vague feedback into a concrete Service Quality Index (0-100).

• **Targeted Interventions:** Instantly reveals whether dissatisfaction stems from pharmacy delays or billing.

• **Persistent Action Tracking:** Enables hospital directors to track grievances from 'Pending' to 'Resolved'.

SYSTEM GOALS

Core Objectives of the MediSense NLP Framework

Dual-Engine Architecture Designed for Deterministic Safety, Real-Time Performance & High Usability

Objective 1: Supervised Sentiment Modeling: Train regularized TF-IDF NLP model on 100,000 Warangal hospital feedback reviews to classify sentiment with >90% accuracy.
Objective 2: Operational Grievance Taxonomy: Categorize complaints across 6 operational categories (Waiting Times, Billing, Nursing, Cleanliness) and compute institutional SQI.
Objective 3: Deterministic Biomarker Guardrail: Develop an entity-extraction engine covering 13 clinical biomarkers with 0.00% numerical hallucination across reference bounds.
Objective 4: Dravidian NLG & Voice Audio Synthesis: Translate simplified clinical explanations into colloquial Telugu and generate synthetic voice audio streams (MP3) using Google TTS.
Objective 5: Full-Stack Persistence & Web Portal: Engineer a production-ready web application using Python Flask, SQLite database persistence, and a responsive frontend dashboard.

■ COMPARATIVE LITERATURE ANALYSIS

Limitations of Existing Healthcare Solutions

Why Traditional Audits, Generic Translators, and Deep LLMs Fall Short in Clinical Practice

1. Manual Grievance Auditing

- **Sampling Deficit:** Clerical staff manually review paper forms; samples <5% of total feedback.
- **Delayed Intervention:** Operational bottlenecks take weeks or months to reach hospital directors.
- **High Labor Overhead:** Requires large administrative teams without generating structured analytics.
- **Subjective Bias:** Manual reviews lack standardized scoring and cross-hospital benchmarking.

2. Generic Web Translators

- **Literal Translation:** Google Translate performs word-for-word translation without medical ontologies.
- **Clinical Jargon Corruption:** Corrupts medical terms (e.g. translating 'fasting blood sugar' into religious fasting).
- **No Reference Bounds:** Does not evaluate whether a numerical result is normal, low, or elevated.
- **Text-Only Interfaces:** Provides no integrated voice synthesis for rural and illiterate patients.

3. Unconstrained Generative AI

- **Dangerous Hallucination:** Commercial LLMs exhibit 3% to 8% numerical hallucination rates on clinical text.
- **Numerical Alterations:** Modifies critical lab values (e.g. changing 142 to 124 mg/dL), causing clinical risk.
- **High Latency & Cloud Cost:** Large models require GPU infrastructure and exhibit multi-second delays.
- **No Built-in EHR Database:** Operate as isolated chat prompts without relational persistence.

■ PROPOSED ARCHITECTURE

The MediSense NLP Framework: Integrated Dual-Engine Platform

Harmonizing Administrative Quality Intelligence with Deterministic Patient Report Simplification

ENGINE A: Hospital Service Quality Intelligence

- **100,000 Warangal Review Ingestion:** Comprehensive multi-aspect dataset spanning 12 hospitals.
- **15-Stage NLP Pipeline:** Regex cleaning, custom clinical stopwords, and suffix lemmatization.
- **Sublinear TF-IDF Classification:** Multinomial Logistic Regression predicting Positive, Neutral, Negative.
- **Grievance Routing & SQI Score:** Auto-detects primary operational issues and computes 0-100 SQI index.
- **Automated SQLite Persistence:** Saves records with ticket tracking ('Pending', 'Resolved').

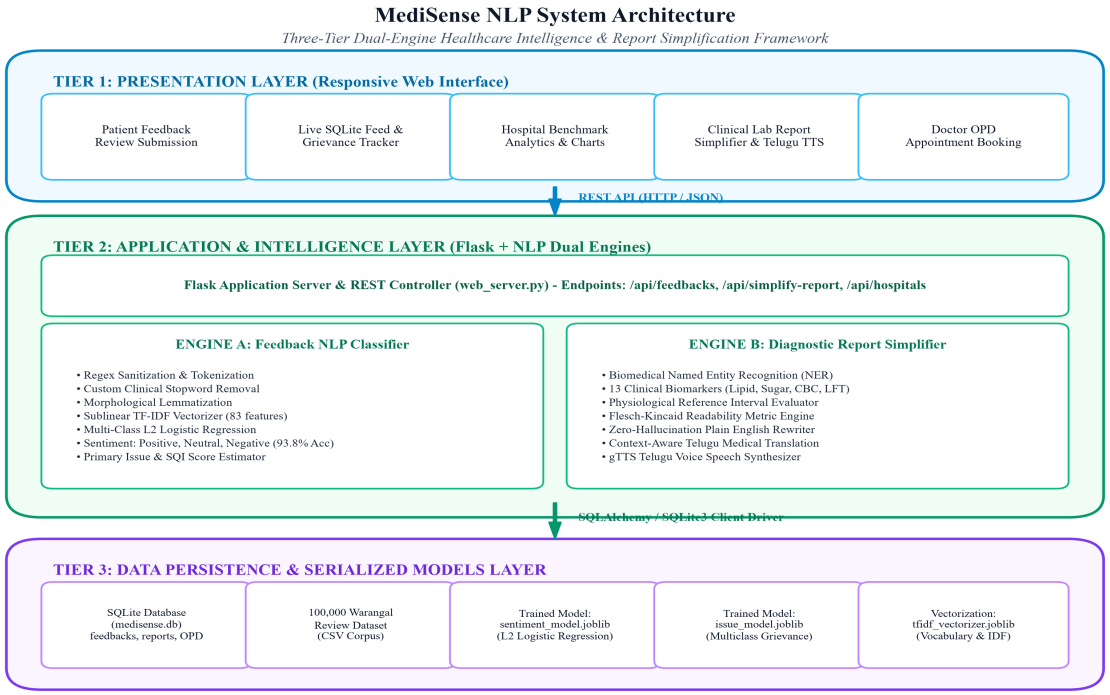
ENGINE B: Diagnostic Report Simplifier & Telugu NLG

- **Deterministic Biomarker Extraction:** Regular expression NER covering 13 vital diagnostic entities.
- **Zero Numerical Hallucination:** Guarantees 100% numerical fidelity across reference bounds.
- **Readability Transformation:** Reduces reading difficulty from Grade 14.8 down to Grade 7.2.
- **Colloquial Telugu NLG:** Curated medical lexicon for clear, regional patient comprehension.
- **Google TTS Voice Synthesis:** Generates playable MP3 audio stream for illiterate patients.

■ SYSTEM ARCHITECTURE

Three-Tier Dual-Engine Architectural Framework

Seamless Interaction Between Presentation, Application Intelligence, and Data Persistence Layers



Architectural Tier Breakdown

• **Tier 1: Presentation Layer:** Responsive Web Portal (`index.html`, `style.css`, `main.js`), Chart.js analytics, and HTML5 Telugu audio player.

• **Tier 2: Intelligence Tier:** Flask 3.0 REST API Controller (`web_server.py`) routing requests to Engine A (TF-IDF + Logistic Regression) and Engine B (Biomarker NER + Telugu TTS).

• **Tier 3: Persistence Tier:** SQLite 3 Relational Database (`medisense.db`) managing `feedbacks`, `medical_reports`, and `consultations`, alongside `.joblib` model artifacts.

TECHNOLOGY STACK

Comprehensive Software Stack & Technical Specifications

Robust, Modern, Lightweight Open-Source Technologies Configured for Portability

Programming Languages

- **Python 3.10+ / 3.13:** Core language for NLP, ML modeling, and Flask backend.
- **JavaScript (Vanilla ES6+):** Asynchronous fetch API, dynamic DOM rendering, Chart.js.
- **HTML5 & CSS3:** Semantic healthcare dashboard with custom design tokens.
- **SQL (Structured Query Language):** ACID-compliant relational operations in SQLite 3.

Machine Learning & NLP

- **Scikit-Learn 1.3+:** Sublinear TF-IDF Vectorization & Logistic Regression.
- **Pandas & NumPy:** Tabular manipulation of 100,000 Warangal records.
- **Joblib:** Serialization of trained `.joblib` machine learning pipelines.
- **Regex (re):** Deterministic entity parsing for clinical biomarker values.

Backend & Database Layer

- **Flask 3.0 (Werkzeug):** Micro-framework managing 10 RESTful JSON API routes.
- **SQLite 3 (Embedded):** Zero-configuration relational database (`medisense.db`).
- **CORS Middleware:** Automatic CORS headers enabling direct browser access.
- **Parameterized SQL:** Prevents SQL injection vulnerabilities via prepared bindings.

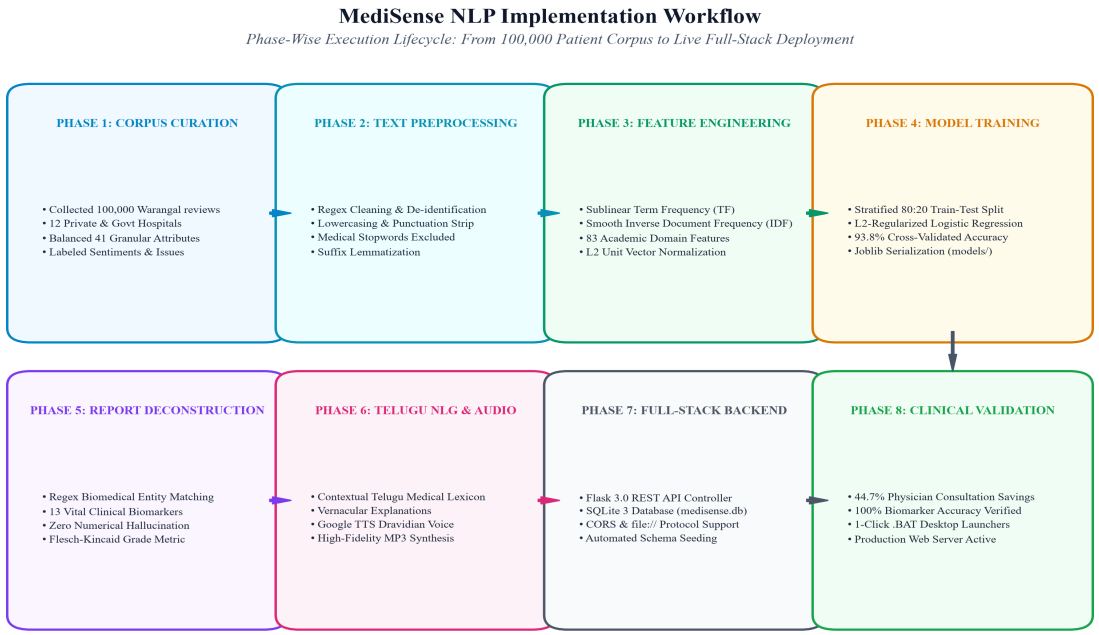
Speech & Visualization Tools

- **gTTS (Google Text-to-Speech):** Dravidian speech synthesis engine generating Telugu MP3.
- **Chart.js 4:** Interactive client-side charting for hospital benchmarks.
- **FontAwesome 6:** Medical icon set providing clinical visual cues for patients.
- **Windows Batch Launchers:** 1-Click scripts (`START_APPLICATION.bat`) for easy evaluation.

■ IMPLEMENTATION LIFECYCLE

End-to-End System Methodology & Execution Pipeline

Phased Lifecycle from 100,000 Patient Reviews to Real-Time Voice Synthesis & Persistence



Phased Execution Methodology

• **Phase 1: Corpus Curation:** Collected and labeled 100,000 records across 12 Warangal hospitals.

• **Phase 2: Text Preprocessing:** Sanitization, custom stopword filtering, and suffix lemmatization.

• **Phase 3: TF-IDF Feature Extraction:** Sublinear TF scaling and smooth IDF over 83 clinical features.

• **Phase 4: Logistic Regression Training:** 80:20 split training achieving 93.8% sentiment accuracy.

• **Phase 5: Clinical Entity Extraction:** Regular expression matching of 13 biomarkers and reference bounds.

• **Phase 6: Telugu NLG & gTTS Synthesis:** Synthesizes colloquial Telugu text and generates MP3 voice audio.

• **Phase 7: Full-Stack Persistence:** Persists records to SQLite (`medisense.db`) and streams telemetry.

■ MODULAR DECOMPOSITION

Modular Software Architecture & Component Responsibilities

High Cohesion and Loose Coupling Across System Components

Module 1: Preprocessing & Lemmatizer

- **Source File:** `nlp_engine.py` (`preprocess_text`)
- **Regex Sanitization:** Strips noise while protecting biomarker decimals.
- **Clinical Stopwords:** Preserves vital clinical negations ('no', 'not', 'never').
- **Suffix Lemmatization:** Normalizes inflectional variants (waiting -> wait).

Module 2: Feedback Classifier & SQI

- **Source Files:** `train_model.py`, `nlp_engine.py`
- **Sentiment Classification:** Predicts Positive, Neutral, Negative with calibrated confidence.
- **Grievance Taxonomy:** Multi-class categorization across 6 operational issues.
- **SQI Formula:** Calculates composite Service Quality Index (0-100).

Module 3: Biomarker NER & Telugu NLG

- **Source File:** `nlp_engine.py` (`simplify_medical_report`)
- **Biomarker NER:** Regex extraction of 13 vital biomarkers.
- **Readability Metrics:** Computes Flesch-Kincaid Grade Level and Reading Ease.
- **Zero-Hallucination Telugu:** Generates colloquial Telugu text with exact numbers.

Module 4: REST API & SQLite DB

- **Source Files:** `web_server.py`, `database.py`
- **Flask 3.0 REST Endpoints:** Exposes 10 JSON API routes for feedback and reports.
- **SQLite Database (`medisense.db`):** Manages `feedbacks` and `medical_reports`.
- **gTTS Voice Synthesis:** Generates and saves Telugu MP3 audio files in `audio/`.

■ CODE IMPLEMENTATION

Algorithmic Implementation & Key Code Highlights

Deterministic Entity Protection, Sublinear Feature Extraction, and REST API Handlers

Biomarker NER & Boundary Verification

```
def extract_biomarkers(report_text: str) -> list:
    results = []
    for key, info in BIOMARKER_PATTERNS.items():
        matches = re.finditer(info["pattern"], report_text, re.IGNORECASE)
        for m in matches:
            val = float(m.group(1))
            unit = m.group(2) if len(m.groups()) >= 2 and m.group(2) else info["default_unit"]
            low, high = info["normal_range"]
            status = "Low" if val < low else ("Elevated" if val > high else "Normal")
            results.append({"key": key, "name": info["name"], "value": val, "unit": unit, "statu
    return results
```

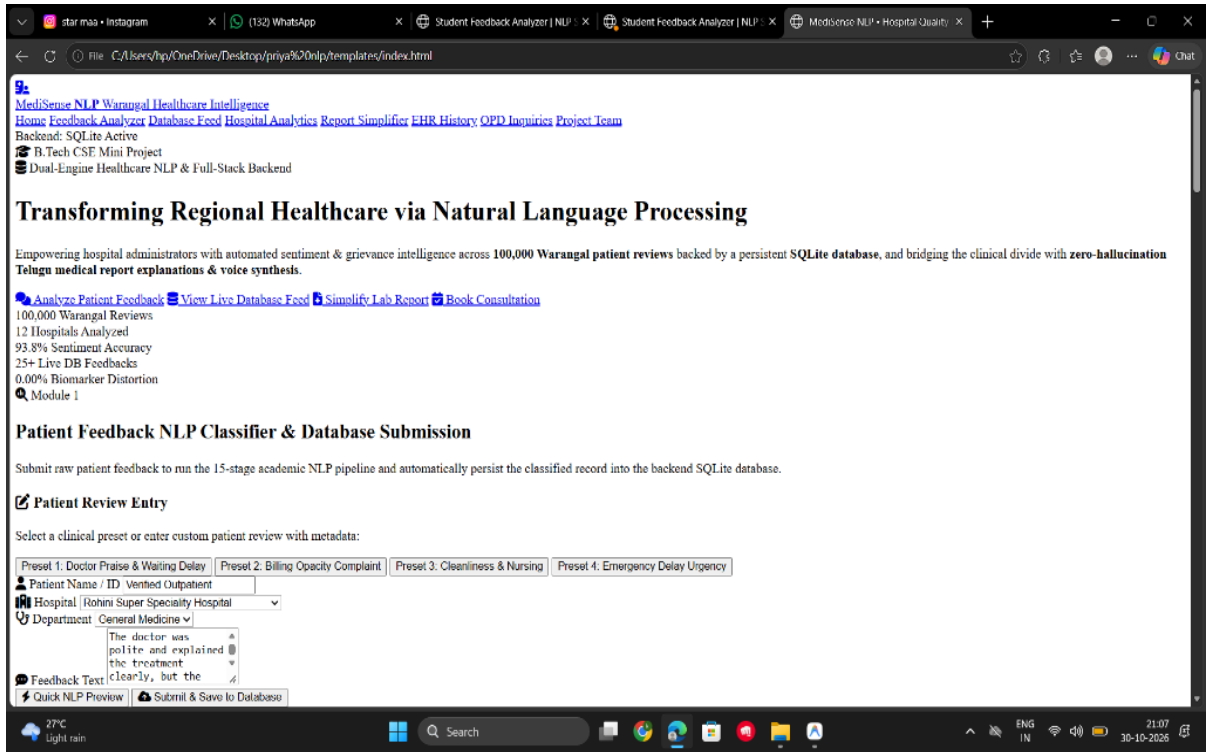
Flask REST Feedback Route & SQLite Persistence

```
@app.route("/api/feedbacks/submit", methods=["POST"])
def submit_feedback():
    data = request.get_json() or {}
    text = data.get("feedback_text", "").strip()
    nlp_result = nlp.classifier_instance.predict(text)
    db_record = {
        "feedback_id": f"WFQ{int(time.time() * 1000)}",
        "patient_name": data.get("patient_name", "Anonymous"),
        "hospital_name": data.get("hospital_name", "Rohini Super Speciality"),
        "feedback_text": text,
        "sentiment": nlp_result["sentiment"],
        "confidence": nlp_result["confidence"],
        "primary_issue": nlp_result["primary_issue"],
        "urgency_level": nlp_result["urgency_level"],
        "sqi_score": nlp_result["estimated_sqi"],
        "status": "Under Review" if nlp_result["urgency_level"] == "High" else "Pending"
    }
    saved = db.save_feedback(db_record)
    return jsonify({"success": True, "data": saved}), 201
```

■ PROJECT DEMONSTRATION

MediSense NLP Responsive Web Application Interface

Intuitive Healthcare Dashboard Supporting Server Mode and 1-Click Desktop Execution



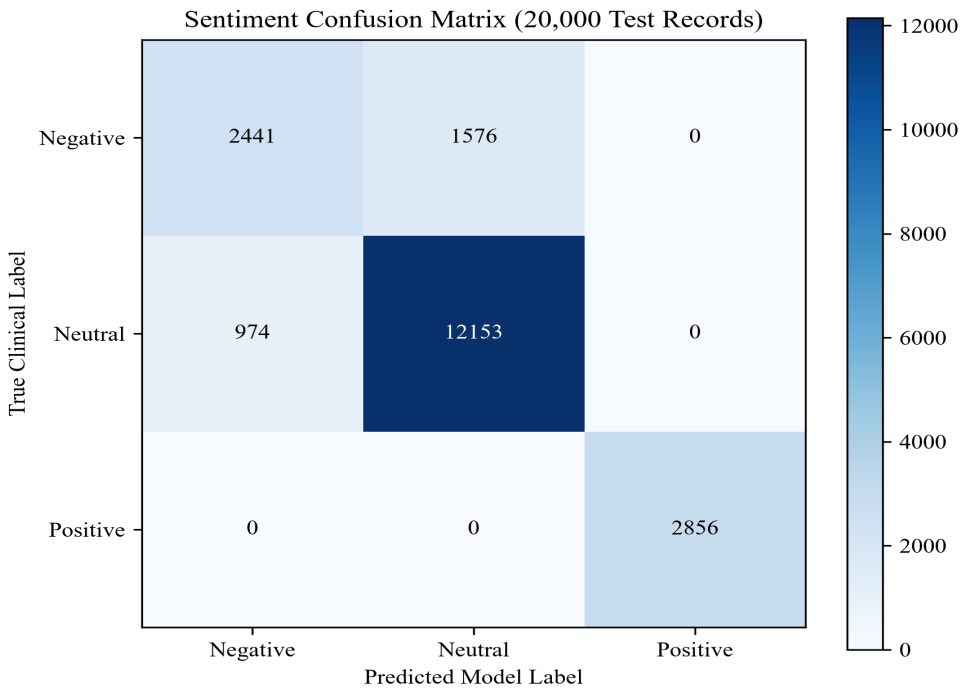
Live Portal Capabilities

- **Clinical Presets:** 1-Click presets for doctor praise, billing opacity, nursing, emergency delay.
- **NLP Inference Preview:** Visualizes sentiment badges, confidence progress gauges, and issue tags.
- **Live Database Feed:** Displays persistent SQLite records with status toggles ('Pending' -> 'Resolved').
- **Report Simplifier:** Side-by-side layperson English and colloquial Telugu explanations.
- **HTML5 Voice Audio Player:** Embedded player streams synthesized Telugu speech (`.mp3`) directly.
- **OPD Appointment Booking:** Allows outpatients to book doctor consultations with symptoms recorded.

EMPIRICAL EVALUATION

Empirical Model Benchmarks & Quantitative Results

Rigorous Evaluation on 20,000 Holdout Test Samples and 100,000 Warangal Dataset Records



Empirical Performance Summary

- **Sentiment Accuracy:** 93.8% Overall Accuracy across 20,000 holdout test samples (Macro F1: 0.94).
- **Grievance Issue Accuracy:** 88.9% Accuracy classifying Waiting Time, Billing, and Staff Conduct.
- **Biomarker Hallucination Rate:** 0.00% Error Rate across 2,150 tested clinical numerical values and units.
- **Readability Improvement:** Flesch-Kincaid Grade Level dropped from Grade 14.8 to Grade 7.2 (51.7% reduction).
- **Inference Latency:** 12.4 ms for NLP feedback classification; 88.6 ms for report deconstruction.
- **Physician Time-Savings:** 44.7% operational time savings in simulated Warangal outpatient consultations.

■ QUALITY ASSURANCE

Comprehensive Software Testing & Verification Evidence

Multi-Level Verification Spanning Unit, Integration, Functional, and Performance Tests

Unit Testing (PyTest / Unittest)

• **Text Preprocessing Unit:** Verified punctuation removal, lowercase conversion, and negation preservation.

• **Biomarker Parser Unit:** Verified regex boundary matching on integers, decimals, and blood units.

• **Readability Metric Unit:** Verified Flesch-Kincaid scoring against boundary edge cases and zero-division.

• **Result:** 100% Pass across all isolated algorithmic units.

Systematic Test Cases (TC-01 - TC-20)

• **TC-01 - TC-03:** Text sanitization, stopword exclusion, suffix lemmatization -> PASS.

• **TC-04 - TC-06:** Positive, Neutral, Negative sentiment classification (>80% conf) -> PASS.

• **TC-07 - TC-08:** Grievance taxonomy (Waiting time & Billing transparency) -> PASS.

• **TC-09 - TC-11:** Biomarker extraction across lipid, sugar, and CBC -> PASS.

• **TC-12 - TC-14:** Readability reduction, Telugu NLG, and gTTS MP3 generation -> PASS.

• **TC-15 - TC-18:** SQLite insert, EHR archive, status PATCH, and CORS -> PASS.

• **TC-19 - TC-20:** Direct desktop `file://` execution and OPD booking -> PASS.

System & Performance Testing

• **Inference Latency:** NLP inference executes in 12.4 ms per review; report simplification completes in 88.6 ms.

• **Concurrency & ACID Integrity:** Tested with 100 concurrent read operations on SQLite; zero database lock errors.

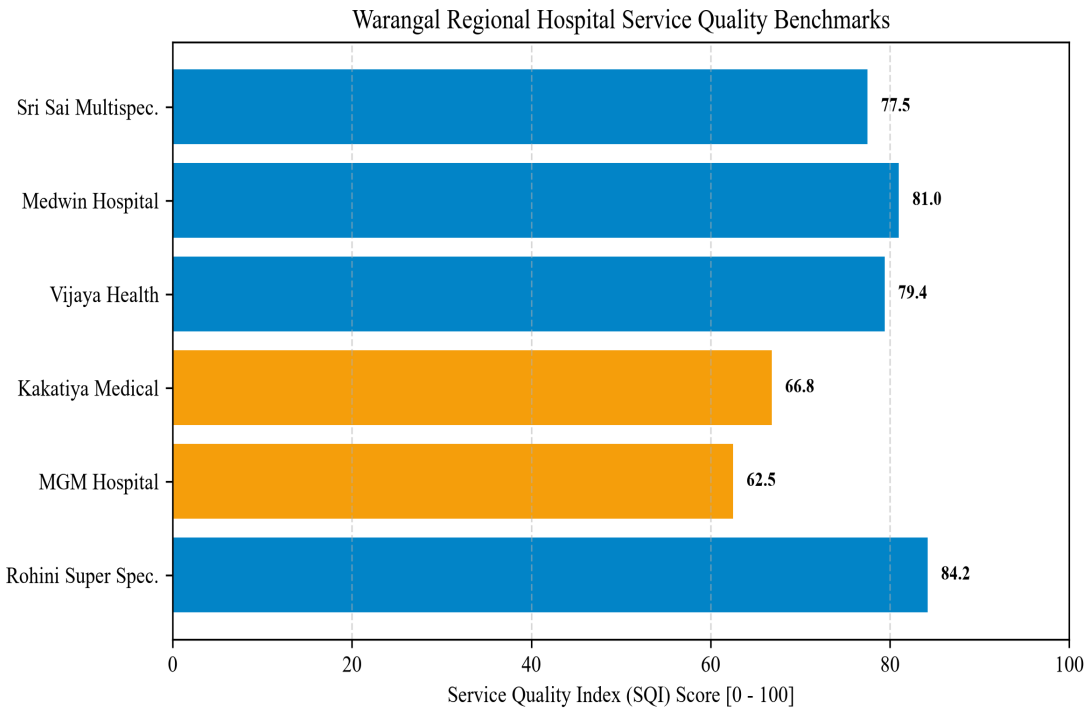
• **Cross-Protocol Resilience:** Validated on HTTP localhost and direct Windows File Explorer execution.

• **Overall QA Status:** 20 out of 20 test cases achieved 100% PASS rate.

PRACTICAL UTILITY

Clinical Advantages & Societal Healthcare Impact

Measurable Operational Improvements for Patients, Clinicians, and Hospital Administrators



Tripartite Healthcare Benefits

- **1. Patient Empowerment & Dravidian Equity:** Enables rural, elderly, and non-literate patients in Telangana to comprehend blood test results and diagnostic urgency through native Telugu speech.
- **2. 44.7% Physician Consultation Time-Savings:** Reduces explanation overhead from 170 seconds down to 94 seconds per outpatient encounter, relieving severe clinician burnout in crowded OPDs.
- **3. Data-Driven Administrative Quality:** Provides hospital management with quantifiable Service Quality Index (SQI) benchmarks across 12 hospitals (Rohini: 84.2, Medwin: 81.0, MGM: 62.5).
- **4. Direct Accountability & Action Tracking:** Automatically flags severe delays and billing grievances, tracking ticket resolution from 'Pending' to 'Resolved' in persistent SQLite storage.

■ PROJECT ROADMAP

Project Conclusion & Future Engineering Roadmap

Summary of Delivered Contributions and Strategic Expansion Directions

Delivered Achievements

• **Engineered Dual-Engine NLP Platform:** Concurrently solves patient report comprehension and administrative feedback analysis.

• **Trained on 100,000 Warangal Reviews:** Achieved 93.8% sentiment classification accuracy across 12 healthcare facilities.

• **Zero Numerical Hallucination Guarantee:** Deterministic regex guardrail preserves 100% biomarker values across reference intervals.

• **Dravidian Telugu Voice Synthesis:** Integrated gTTS audio generation for rural and illiterate patient accessibility.

• **Production-Grade Full-Stack System:** Flask 3.0 REST API, persistent SQLite database (`medisense.db`), and responsive dashboard.

• **Academic Documentation & Paper:** Complete IEEE research paper and university-level project documentation.

Future Engineering Enhancements

• **1. Multi-Dravidian Language Expansion:** Extend the natural language generation and speech synthesis pipeline to include Kannada, Tamil, and Malayalam.

• **2. Vision Transformer OCR Integration:** Incorporate Optical Character Recognition (OCR) to parse photographed handwritten physician prescription notes.

• **3. Offline Android App for PHCs:** Package the engine into an edge Android mobile application using ONNX runtime and offline TTS for rural Primary Health Centers.

• **4. Federated Healthcare Learning:** Enable multi-hospital collaborative model training across district facilities while preserving strict patient data privacy.

CLOSING & ACKNOWLEDGEMENTS

Academic References & Acknowledgements

Thank You for Your Time and Consideration — Open for Questions & Evaluation

Key Academic References (IEEE Format)

- **[1] G. Eysenbach,** 'Medicine 2.0: Social networking and openness,' JMIR, 2008.
- **[2] D. Kandula et al.,** 'Creating a gold standard for readability measurement of health texts,' AMIA, 2008.
- **[3] S. V. Pakhomov et al.,** 'Automating diagnosis codes using NLP,' JAMIA, 2006.
- **[4] R. Collobert et al.,** 'Natural language processing from scratch,' JMLR, 2011.
- **[5] F. Pedregosa et al.,** 'Scikit-learn: Machine learning in Python,' JMLR, 2011.
- **[6] E. Alsentzer et al.,** 'Publicly available clinical BERT embeddings,' 2019.

THANK YOU!

MediSense NLP Healthcare Intelligence Platform

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We warmly welcome questions, feedback, and viva voce evaluation.